Phenoscaping Reveals Multimodal γδ T-cell Cytotoxicity as a Strategy to Overcome Cancer Cell–Mediated Immunomodulation

Abstract

γδ T cells can kill cancer cells via antibody-independent cytotoxicity (AIC) and antibody-dependent cellular cytotoxicity (ADCC). A better understanding of how these cytotoxic mechanisms are affected by different cancer cells and different T-cell donors could help identify improved immunotherapeutic strategies. To test the combinatorial interactions among T cell interdonor heterogeneity, cancer cell intertumor heterogeneity (ITH), and multimodal γδ T-cell killing, we performed a systematic single-cell phenoscaping analysis of more than 1,000 γδ T-cell and colorectal cancer patient-derived organoid cultures. Single-cell analysis of posttranslational modification (PTM) signaling, cell cycle, apoptosis, and T-cell immunophenotypes revealed that whereas unmodified γδ T cells have limited antitumor activity, IL15Rα–IL15 fusion protein [stabilized IL15 (stIL15)]-engineered γδ T cells can kill patient-derived organoids via AIC without exogenous cytokine support. However, when stIL15 γδ T cells only killed via AIC, cancer cells reciprocally rewired γδ T-cell PTM signal networks in an ITH-specific manner to suppress anticancer cytotoxicity. stIL15 γδ T cells could overcome this cancer cell immunomodulation by also engaging B7-H3–targeted ADCC independent of B7-H3 checkpoint activity. Combined AIC and ADCC rescued γδ T-cell PTM signaling flux and enabled γδ T cells to kill chemorefactory revival colon cancer stem cells. Together, these results demonstrate that multimodal γδ T-cell cytoxicity mechanisms can overcome ITH-specific immunomodulation to kill chemorefractory cancer cells.

Significance:

Single-cell phenoscaping of more than 1,000 γδ T-cell and patient-derived organoid cultures shows that cancer cells suppress anticancer γδ T-cell cytotoxicity but γδ T cells can use multimodal killing to overcome immunomodulation.

Introduction

Adoptive T-cell therapies are an important new class of anticancer immunotherapies (1). Despite the success of αβ chimeric antigen receptor T (CAR-T) cells for the treatment of hematologic malignancies, the efficacy of cellular therapies against solid tumors remains limited (2). Furthermore, the reliance of CAR-T cells on a single killing modality can limit their cytotoxic potential against antigen-heterogeneous cancers (3). Alternative adoptive cellular therapies based on γδ T cells and NK cells are undergoing evaluation as anticancer biotherapeutics, but our understanding of how these cells are regulated by genetic engineering and interactions with heterogenous cancer remains limited.

Vγ9Vδ2 γδ T cells (hereafter γδ T cells) are a subset of cytotoxic T cells that can kill cancer cells via multiple mechanisms (4). Antibody-independent cytotoxicity (AIC) is achieved by both MHC-independent function of the γδ T-cell receptor (TCR) and NK-like receptors such as NKG2D, DNAM-1, and the natural cytotoxicity receptors NKp30, NKp44, and NKp46 (5). Additionally, γδ T cells can perform antibody-dependent cellular cytotoxicity (ADCC) via Fcγ receptor CD16 (FcγRIII) and tumor-bound IgG (6–8). γδ T cells can be genetically engineered (9, 8), are safe in the allogeneic setting (10), and can be readily expanded for clinical use (4). The various γδ T-cell killing modalities, as well as their ability to persist within and serially kill solid tumors, can be further enhanced via engineered secretion of common γ-chain cytokines (9, 8).

Colorectal cancer is a solid tumor of the colonic and rectal epithelium that results in the death of >900,000 people per year (11). Colorectal cancer tumors are complex heterocellular systems that possess substantial nongenetic plasticity (12) and an immunosuppressive tumor microenvironment (TME; ref. 13). Although ~15% of patients display microsatellite instability (MSI) leading to high tumor mutational burden (TMB) and remarkable neoadjuvant immune-checkpoint blockade (ICB) responses (14), the majority of patients with colorectal cancer have microsatellite stable (MSS) disease with a low TMB and respond poorly to immunotherapies (15). Furthermore, colorectal cancer cells possess high phenotypic plasticity (16, 17) that can drive nongenetic resistance to standard-of-care chemotherapies via stem cell transdetermination from chemosensitive proliferative colonic stem cells (proCSC) to chemorefractory revival CSCs (revCSC; refs. 12, 18). Consequently, there is a desperate need to develop new treatment options for chemoresistant revCSC-dominant MSS colorectal cancer.

A primary challenge in the treatment of cancer is the large variance in phenotype and treatment response between patients—commonly known as “intertumor heterogeneity” (ITH; ref. 19). In the context of allogeneic cellular therapies, the complexity of ITH is further compounded by variability between cell products derived from different peripheral blood mononuclear cell (PBMC) donors—so-called “interdonor heterogeneity” (IDH; refs. 20, 21). How cancer ITH interacts with γδ T-cell IDH across a range of T-cell killing mechanisms is unknown.

Phenoscaping is an emerging experimental technique that combines systematic high-throughput perturbations with high-dimensional single-cell analysis to map the phenotypic regulators of a dynamic cellular system (22). We recently used phenoscaping to map the cell-intrinsic and -extrinsic regulators of CSC plasticity (23), but to date, no phenoscaping study of T-cell–cancer cell interactions has been reported.

In this study, we present a large-scale single-cell phenoscaping analysis of γδ T-cell interactions with patient-derived organoid (PDO) models of colorectal cancer to chart the regulators of multimodal γδ T-cell cytotoxicity across cancer ITH and T-cell IDH. Using thiolreactive organoid barcoding *in situ* mass cytometry (TOB*is* MC; refs. 24, 25), we studied the posttranslational modification (PTM) signaling, cell state, apoptosis, and immunophenotype of γδ T-cell–PDO interactions in 3D across >1,000 perturbations at single-cell resolution.

Phenoscaping analysis revealed that unmodified γδ T cells have variable expansion phenotypes and poor cytotoxicity against colorectal cancer PDOs. However, γδ T cells engineered to express an IL15Rα–IL15 fusion protein (stabilized IL15, stIL15; refs. 8, 26) are cytotoxic against colorectal cancer PDOs but can be reciprocally immunomodulated by PDOs in an ITH-specific manner, including global rewiring of γδ T-cell PTM signaling networks. We find that tumor-derived γδ T-cell immunomodulation can limit anticancer killing when γδ T cells use AIC alone but that immunomodulation can be overcome by cancer cell opsonization using anti–B7-H3 mAb therapy, which engages γδ T-cell ADCC capacity. Crucially, we find that engineered γδ T cells can kill MSS PDOs that are enriched for chemorefractory revCSCs. Taken together, these results demonstrate the value of high-throughput cellular therapy phenoscaping and reveal that γδ T-cell multimodal cytotoxicity can overcome immunomodulation to kill chemoresistant cancer cells.

Materials and Methods

Organoid culture

Colorectal cancer PDOs were obtained from the Human Cancer Models Initiative with written informed consent (Sanger Institute; ref. 27) and expanded in 3 × 25 μL droplets of Growth Factor Reduced Matrigel (Corning, 354230) per well of a 12-well plate (Helena Biosciences, 92412T). Each well was supplemented with 1 mL of PDO expansion media comprising Advanced DMEM F/12 (Thermo Fisher Scientific, 12634010) containing 2 mmol/L L-glutamine (Thermo Fisher Scientific, 25030081), 1 mmol/L N-acetyl-L-cysteine (Sigma, A9165), 10 mmol/L HEPES (Sigma, H3375), 500 nmol/L A83-01 (Generon, 04–0014), 10 mmol/L SB202190 (Avantor, CAYM10010399-10), 1X B-27 Supplement (Thermo Fisher Scientific, 17504044), 1X N-2 Supplement (Thermo Fisher Scientific, 17502048), 50 ng mL^{-1} EGF (Thermo Fisher Scientific, PMG8041), 10 nmol/L gastrin I (Sigma, SCP0152), 10 mmol/L nicotinamide (Sigma, N0636), 1X HyClone penicillin–streptomycin solution (Thermo Fisher Scientific, SV30010), and conditioned media produced as described in ref. 28 at 5% CO_2, 37°C. L-cells for conditioned media production were obtained from Shintaro Sato (Research Institute of Microbial Diseases, Osaka University; ref. 29). PDOs were dissociated into single cells with 1X TrypLE Express Enzyme (Gibco, 12604013; incubated at 37°C for 20 minutes) and passaged every 5 to 10 days. For the first 24 hours after dissociation, media were also supplemented with 10 μmol/L Rho-associated protein kinase inhibitor (ROCKi; Y-27632, Sigma, Y0503). L-cells for conditioned media production were obtained from Shintaro Sato (Research Institute of Microbial Diseases, Osaka University). To aid cell type–specific visualization and gating, colorectal cancer PDOs were transfected with H2B-RFP (Addgene, 26001).

Generation of CRISPR B7-H3KO PDOs

A custom lentivirus containing a GFP reporter, Cas9, and guide RNA against B7-H3 (CTG GTGCACAGCTTTGCTG) was ordered from Merck. PDOs were dissociated into single cells with 1X TrypLE Express Enzyme (Gibco, 12604013; incubated at 37°C for 20 minutes). Single-cell PDOs were transduced at an multiplicity of infection of 5 in PDO expansion media supplemented with 10 μmol/L ROCKi and 8 μg/mL polybrene (Merck, TR-1003-G) with spinoculation at 300 x *g* for 1 hour at room temperature. Following transduction, single-cell PDOs were passaged and expanded as normal. Upon reaching a sufficient number for cell sorting, PDOs were dissociated into single cells and washed in FACS buffer [PBS (Gibco) supplemented with 1 mmol/L (Sigma, 03690), 25 mmol/L HEPES, and 1% FBS]. Cells were stained with 1 μg/mL of anti-human B7-H3 primary antibody (Prof. Kerry Chester, UCL Cancer Institute, London, United Kingdom) and 1 μg/mL of goat anti-human IgG secondary antibody (Thermo Fisher Scientific, A-21445). Cells negative for B7-H3 were bulk sorted through a BD FACSAria III into collection media (50% PDO expansion media and 50% FBS supplemented with 10 μmol/L ROCKi) and grown as normal. PDOs underwent a repeat bulk sort following multiple passages to select a stable B7-H3KO population. 100% B7-H3KO efficiency of the twice-sorted PDO population was confirmed with B7-H3 gene amplification and Sanger sequencing.

PBMC isolation

This study was approved by a national Ethics Committee (West Midlands HRA, 14/WM/1253) and complied with the World Medical Association Declaration of Helsinki with written informed consent. Leukopaks from healthy donors were purchased from Cambridge Bioscience, and PBMCs were isolated by density-adjusted centrifugation using Lymphopure density gradient medium (BioLegend, 426201). Residual red blood cells were removed using ammonium–chloride–pottasium (ACK) lysing buffer (Thermo Fisher Scientific, A1049201), and platelets were removed with slow-speed centrifugation. PBMCs were immediately frozen in a freezing solution containing 10% DMSO, with storage complying with the Human Tissue Act 2004. Owing to the size and longitudinal nature of the screen, cryopreserved PBMCs were used so that a consistent bank of γδ T cell donors could be utilized across multiple batches of experiments.

γδ T-cell expansion

Cryopreserved PBMCs were thawed and rested overnight in RPMI 1640 medium supplemented with 1X GlutaMAX and 10% (v/v) FBS (Gibco, 10082147) at 5% CO2, 37°C. For Vγ9Vδ2 expansion, overnight-rested PBMCs were plated in tissue culture–treated plates at 2 × 106 cells/cm2 in fresh media and stimulated with 5 μmol/L zoledronic acid (Actavis) and IL2 (100 IU/mL; aldesleukin, Novartis). Unless otherwise stated, IL2 was replenished every 2 to 3 days by removing 50% of the media from the well and replacing with fresh media containing IL2 (200 IU/mL). γδs were expanded until day 12 at which point they were harvested for experimental use and flow cytometry immunophenotyping. By day 12, γδ T cells typically represent the predominant cell type, with remaining populations of NK cells and *αβ* T cells varying by donor.

Lentiviral vectors

Gene constructs containing both enhanced GFP (eGFP) and stIL15 were designed using SnapGene 2.8.3. Gene blocks were synthezised by Twist Bioscience and cloned into a pCCL.SFFV lentiviral vector using standard restriction enzyme cloning. eGFP–stIL15 lentivirus was produced in-house and was titred on HEK293T cells (CVCL_0063) with the number of titratable U/mL determined by flow cytometry.

Lentiviral transduction of γδ T cells

Following 48 hours of zoledronic acid and IL2 stimulation, lentivirus was diluted in Opti-MEM (Thermo Fisher Scientific, 31985062) and added directly to the γδ T-cell culture wells at a multiplicity of infection of 4. Spinoculation and transduction enhancers were not used for lentiviral transduction. Transduced γδs were expanded until day 12, at which point, they were harvested for experimental use and flow cytometry immunophenotyping. Transduction efficiency was calculated as the percentage of live Vδ2 expressing GFP.

Flow cytometry

Flow cytometric analysis was performed using a FACSymphony A5 flow cytometer (BD Biosciences, SCR_022538). Data were acquired using BD FACSDiva (v8.0.1) and analyzed using FlowJo v10.10.0 and Cytobank (Beckman Coulter). Compensation was calculated with the use of OneComp eBeads (Thermo Fisher Scientific, 01-1111-41) stained with single-color antibodies. Colorectal cancer PDO B7-H3 antigen density was estimated using the BD Quantibrite PE Kit (BD Biosciences, 340495) as per the manufacturer's protocol and a primary anti-human B7-H3 PE antibody (Miltenyi, 130-120-712).

Anti–B7-H3 mAb

The anti–B7-H3 mAb was obtained from a murine single-chain Fv (scFv) phage-display library constructed as previously described (30). B7-H3–binding scFvs were identified by panning and phage monoclonal ELISA. The scFv was subcloned into a chimeric hIgG1 format using a G1m1, 17 heavy chain allotype and produced as a full antibody by Evitria AG (www.evitria.com) using transient expression in CHO cells and protein A purification. Fc-null versions of the same B7-H3 mAb clones were generated via the incorporation of "LALA" mutations into the Fc domain (L234A/L235A) and were also produced by Evitria.